W1 Project Scope: MLOps Pipeline for Medical Model Deployment

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***Abstract*—This document outlines the project scope for the "MLOps Pipeline for Medical Model Deployment" topic. Deploying machine learning models in healthcare presents unique challenges, including data drift, hardware constraints, and strict reliability requirements. This project aims to design an end-to-end industrial pipeline to train, version, deploy, and monitor a medical image classifier model using the ChestMNIST dataset. The pipeline will integrate MLflow for experiment tracking and FastAPI for serving, enabling simulated A/B testing and drift detection while comparing the operational efficiency of two backbone architectures: ResNet-18 and MobileNetV2.**

I. Context and Problem

While modern Machine Learning (ML) techniques have achieved remarkable performance on medical imaging tasks in laboratory settings, translating these successes into clinical practice remains challenging. The primary bottleneck is often the deployment phase, where models are exposed to real-world, out-of-distribution data, changing hardware constraints, and evolving patient populations. A medical image classifier model trained on static data can rapidly degrade in performance due to concept drift. This project focuses on building a robust, production-ready MLOps pipeline specifically for multi-label chest X-ray classification using the lightweight ChestMNIST dataset, demonstrating how to bridge the gap between model training and reliable clinical deployment.

II. Initial State of the Art (SOTA)

The transition from static ML models to Continuous Integration and Continuous Deployment (CI/CD) for ML is a highly active research area. Rather than treating training and deployment as isolated steps, recent advancements emphasize holistic MLOps practices.

Transfer learning for medical imaging: Raghu et al. [1] showed that while ImageNet-pretrained models may not significantly improve final accuracy on medical images compared to training from scratch, they substantially accelerate convergence, making them ideal for rapid MLOps iteration.

Previous works on the dataset: Yang et al. [2], in their introduction of the MedMNIST collection, benchmarked standard architectures on the ChestMNIST dataset. They established baseline AUC scores using standard architectures like ResNet-18 and ResNet-50, highlighting the difficulty of multi-label thoracic disease classification on low-resolution data.

MLOps and Experiment Tracking: In the medical domain, deployment frameworks are increasingly adopting rigorous CI/CD practices. Willemink et al. [3] detailed the necessity of structured MLOps frameworks to facilitate safe clinical AI deployment, emphasizing the need to monitor inference latency and detect distribution shifts in real-time. To support this, Zaharia et al. [4] introduced MLflow as a standard to track hyperparameters, manage model versions, and package code, effectively mitigating the reproducibility crisis in ML engineering. Our project builds upon these SOTA principles by applying an end-to-end operational pipeline to ChestMNIST.

III. Objectives and Proposed Methodology

The core objective of this project is to construct a comprehensive MLOps pipeline that handles the full lifecycle of a transfer-learning-based medical image classifier model.

A. Methodology

- **Experiment Tracking:** We will utilize MLflow to log hyperparameters, metrics, and model artifacts, ensuring full reproducibility across training runs.
- **Model Serving:** The trained models will be deployed using FastAPI, exposing a RESTful endpoint for real-time medical image inference.
- **A/B Testing & Model Comparison:** We will operationalize and compare two distinct backbones—**MobileNetV2** (optimized for speed and low-resource environments) and **ResNet-18** (a standard baseline for accuracy)—in a simulated production environment to evaluate performance trade-offs.
- **Monitoring:** We will simulate an environment where patient data drifts over time and deploy monitoring tools to automatically detect these distribution shifts.

IV. Datasets and Evaluation Metrics

A. Dataset

The project will utilize **ChestMNIST**, a subset of the MedMNIST collection containing multi-label chest radiograph images. Utilizing ChestMNIST ensures highly feasible training times, allowing the team to focus primarily on the engineering, tracking, and operational aspects of the MLOps pipeline rather than computational overhead.

B. Target Metrics

Beyond traditional classification metrics (AUC-ROC, Accuracy, and F1-Score as benchmarked by the dataset creators), the operational efficiency of the deployed models will be heavily evaluated. Key MLOps metrics will include:

- **Inference Latency:** Measured in milliseconds (ms) per request, comparing the serving speed of MobileNetV2 versus ResNet-18.
- **Drift Detection Quality:** The system's ability to accurately and promptly flag simulated data drift in the production environment.

References

[1] M. Raghu, C. Zhang, J. Kleinberg, and S. Bengio, "Transfusion: Understanding Transfer Learning for Medical Imaging," *Advances in Neural Information Processing Systems*, vol. 32, 2019.

[2] J. Yang et al., "MedMNIST v2: A Large-Scale Lightweight Benchmark for 2D and 3D Biomedical Image Classification," *Scientific Data*, vol. 10, no. 1, 2023.

[3] A. Willemink et al., "Medical machine learning operations: a framework to facilitate clinical AI development and deployment in radiology," *European Radiology*, 2024.

[4] M. Zaharia et al., "Accelerating the Machine Learning Lifecycle with MLflow," *IEEE Data Eng. Bull.*, vol. 41, no. 4, pp. 39-45, 2018.